

SAMPREETH AVVARI

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SUMMARY

Forward Deployed Engineer who turns ambiguous, stakeholder-facing problems into shipped software, fluent from the C-suite to production code. Built 5 production ML, LLM, and full-stack systems solo, replacing a \$98K vendor quote in weeks.

EDUCATION

MS, Computer Engineering, New York University, NY (CGPA 3.84/4)

Aug 2023 - May 2025

BTech, Computer Science and Engineering, JNTU, India (CGPA 3.92/4)

Jul 2017 - Jun 2021

SKILLS

Languages & Data: Python, TypeScript, JavaScript, Go, Java, SQL, C++; PostgreSQL, pgvector, MySQL, MongoDB, DynamoDB, Redis, BigQuery

AI / LLM: LLM pipelines, prompt engineering, JSON Schema contracts, RAG (pgvector, Milvus), agentic workflows, evals, fine-tuning, RLHF (GRPO/PPO), QLoRA, PyTorch, Transformers, Vertex AI Gemini, LangChain

Full-stack / Cloud / MLOps: FastAPI, Flask, Node.js, REST, GraphQL, microservices, React, Next.js; GCP (Cloud Run, Pub/Sub), AWS (S3, RDS, MSK), Docker, Kubernetes, Terraform, Airflow, MLflow, Kafka, CI/CD, monitoring (Prometheus, Grafana, OpenTelemetry)

Forward-deployed delivery: stakeholder and customer discovery, scoping, cross-functional collaboration, rapid iteration from user feedback, stakeholder communication, end-to-end ownership, trusted technical advisor; Google Gen AI Leader

WORK EXPERIENCE

Forward Deployed Engineer | Hybridge Implants, New York

Sep 2025 - Present

- Partnered with clinicians, treatment coordinators, and the C-suite, embedded in their workflows to scope ambiguous problems and own delivery end to end, shipping 5 production ML, LLM, and full-stack systems solo.

CBCT Scan Validator, in-house dental-CT quality classifier

- Built an in-house CBCT quality classifier (frozen nnU-Net v2 encoder, OpenVINO, 5.5s/scan) flagging the 10-15% of 600 monthly scans unusable for implant design, replacing a \$98K vendor quote at under \$50 a month.
- Owned the full MLOps loop (SFTP ingest, parity-checked ETL, training, serving) and gated releases on a holdout that caught a leaky 0.80 AUROC, shipping an honest 0.63 with human review.

Treatment Estimator, Next.js and Postgres quoting platform

- Delivered a Next.js and Postgres quoting platform (5-model pricing engine, guided coordinator wizard) now serving 50 consults a week, cutting 20 minutes of manual pricing math per quote to instant.
- Replaced a decade-old vendor tool and hardened its 6-month price guarantee into Postgres database invariants (write-once columns, triggers), covered by 100% branch tests, for under \$35 a month.

Doc Coach, LLM consultation-QA pipeline

- Shipped an LLM consultation-QA pipeline (Cloud Run, FastAPI, Vertex AI Gemini) scoring every doctor consult against a 7-criterion clinical rubric and emailing color-coded coaching reports to the C-suite in 5 minutes.
- Constrained every model response with versioned prompts and JSON Schema contracts, cutting hallucinations 35% under HIPAA-eligible controls while lifting treatment acceptance 130% and downstream revenue 43%.

Software Engineer | Shure Incorporated, India

Aug 2021 - Aug 2023

- Engineered event-driven Flask REST APIs on AWS (DynamoDB, S3, MSK) ingesting 500K+ daily Kafka events, refactoring synchronous calls into async workers to cut processing time 20% at scale.
- Directed delivery across 3 release tracks with Jenkins canary deploys, cutting deployment errors 40% and manual QA 70% with 150+ pytest and Selenium tests, and led code reviews.

RESEARCH EXPERIENCE

Machine Learning Researcher | New York University (writeup)

May 2024 - Sep 2025

- Fine-tuned Llama 3.1 8B through a full RLHF lifecycle (SFT then GRPO/PPO, QLoRA reward models), reaching a 67% win rate over baseline on human evaluation.
- Engineered a Python and Airflow ETL turning 118 Reddit CMV shards into 38K preference pairs with schema enforcement and failure alerting, tracked in WandB (loss, precision, recall, F1).

PROJECTS

JobPilot, open-source (MIT) AI job-hunt autopilot

- Built JobPilot, an open-source job-hunt autopilot adopted by 12 peers, AI-scoring fit and auto-tailoring an ATS-clean resume per match across 11 sources and 39 boards for under \$10 a month.
- Designed 7 agentic patterns: an LLM-as-judge scorer, JSON-schema output contracts, a self-refining rewrite loop, a RAG assistant, multi-step orchestration, tool-use across Gmail, Sheets, and Drive, and guardrails that never auto-send.

Loan Radar, self-retraining loan-default scoring platform

- Architected Loan Radar, a self-retraining loan-default platform on Terraform-provisioned Kubernetes (Ray) with 3 FastAPI microservices, Docker, and MLflow-tracked CI/CD for rollback-safe predictions.